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G H Raisoni International Skill Tech University, Pune

F.Y. B. Tech (AI/AIDS/DS/AIML)
TERM-I (Regular)
CAE I- WINTER 2025 (2024 Pattern)
Foundation of Data Analytics (25UESL1206)

(Time: - 01 Hour) **(Max. Marks: 20)**

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary. iii) Figures to the right indicate full marks. iv) Use of a scientific calculator is allowed. v) Assume suitable data, if necessary.

Q.No	*Question*	Marks	CO	BL
Q1.(a)	Identify and explain the four types of data analytics with a real-world example for each.	[5]	1	1
b)	Differentiate between Data Analysis and Data Analytics based on their focus, scope, and output.	[5]	1	2
	OR			
Q 2.(a)	"Analysis is about diagnosing the current state, while Analytics is about predicting future outcomes." Justify this statement using a suitable example	[5]	1	2
(b)	Define the following buzzwords in the context of data science: i) Internet of Things (IoT) ii) Artificial Intelligence (AI) iii) Machine Learning	[5]	1	1
Q 3.(a)	Compute a) Range, b) Coefficient of Range, c) Quartile Deviation, and Coefficient of Quartile Deviation for the following data: 100, 24, 14, 105, 21, 35, 106, 16, 100,72, 68, 103, 61, 90, 20, 15, 102, 104,	[5]	2	3
(b)	Calculate and interpret the Standard Deviation for the following customer wait times (in minutes): [2, 5, 7, 4, 7]. What does this value indicate about the consistency of the service?	[5]	2	3
	OR			
Q. 4 (a)	A student's semester grades are given below. Calculate the GPA using the weighted mean. Mathematics (4 credits): C (2 points) History (2 credits): A (4 points) Chemistry (3 credits): B (3 points)	[5]	2	3

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	English (2 credits): B (3 points) Lab Work (1 credit): B (3 points)			
(b)	Calculate Variance, Standard Deviation, and Coefficient of Variance for the following Data: 36, 15, 25, 10, 14.	[5]	2	